From Songlines to the Cloud

Episode 1: The Map in Your Head

[Cold open]

To drive a black cab in London, you have to pass a test called the Knowledge. It means learning roughly twenty-five thousand streets within about six miles of Charing Cross, and every route between them, and then proving it to an examiner. It takes years. Not everyone makes it.

In the year 2000, a team of neuroscientists put licensed London cabbies into a brain scanner. They found that the back of the hippocampus, a part of the brain tied to memory and navigation, had significantly more gray matter than in comparable people who didn't drive cabs. And the longer someone had been driving, the bigger the difference.

That raised the obvious question. Were people with big hippocampi drawn to driving cabs? Or did the job build the brain? So the researchers followed seventy-nine trainees through the whole ordeal. The growth showed up only in the ones who qualified. Learning the city had physically changed their brains.

[Beat]

Keep that in mind, because that's where this story really begins. Not in a university, not with the first map, but inside your own skull.

[Theme]

This is *From Songlines to the Cloud*, a history of geography and spatial thinking in six episodes. I'm Paul. I teach history and geography, and over this series I want to tell you a story with a surprisingly clean shape. Spatial thinking starts out as something every human being has. Over thousands of years it gets formalized, measured and professionalized, and in one country in particular, it very nearly gets squeezed out of school altogether. And then, in our own lifetimes, it comes back to everyone.

This first episode is about the beginning: the map in your head, the maps people made before anyone could write, and the first maps that states made once they could.

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Let's start with rats. In 1948 an American psychologist named Edward Tolman, watching rats find their way through mazes, proposed that animals build what he called a "cognitive map": an internal picture that organizes what they know by where things are. It was an idea about the mind. Nobody could point to it.

In 1971, John O'Keefe and Jonathan Dostrovsky pointed to it. Recording from neurons in a rat's hippocampus, they found cells that fired only when the animal was in one particular spot. Move the rat, and a different cell lit up. They called them place cells, and they were the first direct evidence for Tolman's map.

Then in 2005, May-Britt and Edvard Moser found something even stranger next door, in a region called the entorhinal cortex: cells that fire at regular points as an animal moves around. When you plot those points, they form a hexagonal lattice. A grid laid over space. Grid cells, in other words, are an internal coordinate system.

In 2014, O'Keefe and the Mosers shared the Nobel Prize in Physiology or Medicine, "for their discoveries of nerve cells in the brain that enable a sense of place and navigation." Which is to say, a Nobel Prize for the brain's built-in GPS.

Now put the cabbies back in. Spatial thinking isn't just wired in. It responds to use. It's fundamental, *and* it's trainable, which is about as good a description of a school subject as you're ever going to get.

Two things follow from this, and they frame the whole series. First, mapping is older than writing. Long before anyone wrote anything down, people everywhere found ways to store and share what they knew about space, because getting space wrong was how you died. You missed the water. You missed the reef. You missed the pass.

Second, if spatial thinking is a universal human capacity, then the moments when it gets treated as a niche specialty, something for experts, something you can cut from the curriculum, start to look less like the natural order of things and more like an anomaly. We'll get to that anomaly in episode four. For now, let's go to sea.

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The Pacific is the largest ocean on Earth, and people settled it without compasses, without charts, and without writing. They navigated by star paths, by the patterns of ocean swells, by the winds, and by birds. The Marshall Islands alone have been inhabited for something like three thousand years.

For a long time, skeptics didn't buy it. The popular alternative was accidental drift: canoes blown off course, fetching up wherever the current happened to take them. Skill had nothing to do with it.

In 1976 that theory ran into an experiment. The Polynesian Voyaging Society built a double-hulled voyaging canoe called *Hōkūleʻa* and sailed it from Hawai'i to Tahiti, guided by the Micronesian master navigator Mau Piailug, using only traditional methods. No instruments. A man who could read the sky and the sea. They made it. *Hōkūleʻa* has since sailed more than a hundred and fifty thousand nautical miles, including a voyage around the world between 2013 and 2017, and the navigator Nainoa Thompson went on to revive and systematize the old star-compass method for a new generation.

The Marshallese gave this kind of knowledge a physical form, and it's one of the great objects in the history of cartography. They're called stick charts: frameworks of coconut-frond midribs, lashed together with fiber. Shells, or the places where sticks cross, mark islands. The curved sticks mark the dominant ocean swells, and the way those swells bend and interfere with each other as they wrap around an island.

Here's the part I love. These charts generally weren't taken to sea. They were teaching and memory devices. You studied the chart on land, you committed it to memory, and you left it behind. And each chart could be fully read only by the person who made it.

The navigators who carried this knowledge, the *ri-meto*, read the swells by feel, lying in the canoe to sense the motion of the sea. Researchers have since shown that what they were feeling lines up with real oceanographic phenomena. The stick charts fell out of use after the Second World War, with electronic navigation and with the upheaval of Marshallese society, including American nuclear testing at Bikini.

[Beat]

As teachers, I think we should sit with that for a second. A map you study, absorb, and then leave on the beach. The map wasn't the point. The navigator's head was the point.

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Go to the opposite side of the planet, to the east coast of Greenland, and you find another answer to the same problem.

Between 1883 and 1885, the Danish explorer Gustav Holm traveled that coast by umiak, the open skin boat. On the eighth of February 1885, a Tunumiit man named Kunit, from a place called Umivik, sold him a carved block of driftwood. On the twenty-first of March, he sold him another. They're now known as the Ammassalik wooden maps.

They're maps of the coastline, but not like any map you've seen. The coast runs up one side of the block and down the other. Notches mark the fjords and headlands, the places where supplies were stored at old house ruins, and the portage points, where you could haul your boat overland when sea ice blocked the way around a headland.

Now think about the design brief. You're on the water, in the Arctic, quite possibly in the dark. These maps can be read entirely by touch. You can read them with your mittens on. And if you drop one over the side? It floats.

[Beat]

Honestly, I'm not sure anyone has ever beaten that brief. The maps are held today in the Greenland National Museum in Nuuk.

Now go south, to Australia. Aboriginal songlines, sometimes called dreaming tracks, encode routes across the land along the paths of ancestral beings. The knowledge of how to travel is committed to memory in song, which is why the astronomer Ray Norris describes songlines as "oral maps." Documented examples cross the continent. One runs from Heavitree Gap at Alice Springs all the way to Byron Bay on the east coast. Another links the Gulf of Carpentaria to the Snowy Mountains. Star patterns come into it, but as memory aids for the sung route, not as literal charts of the sky.

The memory researcher Lynne Kelly puts it beautifully. Each sacred site along the track anchors part of an information system, so "the knowledge is literally grounded in the landscape."

Song, stick, driftwood. Not one of them is paper, and not one of them needs writing. Every one of them is a survival technology.

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So when do we get the first actual map? This is where I have to put my skeptic's hat on, because the honest answer is: it depends who you ask, and a lot of the famous answers are contested.

Take the cave at Lascaux, in France, painted about seventeen thousand years ago. In the Hall of the Bulls there's a cluster of six dots above an aurochs, and since the 1990s some researchers have argued that those dots are the Pleiades, with other star patterns elsewhere in the cave. It's a lovely idea. Critics point out that the orientations don't match, and nothing else in the archaeological record backs it up. The International Astronomical Union allows it as a possibility, and that's as far as it goes.

There's a term for the underlying problem, and it's too good not to share. *Cartocacoethes.* Roughly: "a mania to see maps everywhere." If you've ever stared at a rock carving and thought, that's a river, that's a field, that's a village, congratulations. You've had cartocacoethes.

The most famous case is a wall painting at Çatalhöyük, in what's now Turkey, from somewhere around 6600 to 6200 BCE. When it was excavated in the 1960s, it was read as a plan of the town's houses, laid out beneath a twin-peaked volcano, Hasan Dağ, in mid-eruption. For decades it was celebrated as the world's oldest map.

Then the researcher Stephanie Meece took it apart. The "volcano," she argued, is better read as a leopard skin, which is a motif all over Çatalhöyük. The "town plan" looks like a common geometric pattern. The volcano's two peaks are the wrong way round compared with the view from the site. The obsidian at Çatalhöyük didn't even come from Hasan Dağ. And if this was a map, where are all the others? There aren't any for thousands of years.

But the story didn't end there. Geologists later dated an eruption of Hasan Dağ to around 6900 BCE, with a margin of error wide enough to overlap the age of the painting. That doesn't prove anyone painted it. It means somebody *could* have seen it.

So where does that leave us? *[Beat]* Every society we know about stored its spatial knowledge in some form, carved or sung or performed. What the deep past can't yet

securely show us is a map in the strict sense. And that means that when the first unambiguous maps do turn up, they turn up somewhere slightly unexpected. Not out on the hunt. In the office.

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In 1882 the British Museum acquired a small clay tablet that had been dug up at Sippar, in Iraq. It's about twelve centimeters by eight, roughly the size of your palm, and it probably dates from somewhere between the late eighth and the sixth century BCE. On it is Babylon, sitting on the Euphrates, inside a circle of land. Around the land runs a ring of ocean called the Bitter River, and beyond that, triangular regions point outward into the unknown.

It's called the Babylonian Map of the World, and it's the oldest known attempt to show the entire known world. It mixes geography with cosmology and myth, and recent work at the British Museum has linked the text on the back to the Mesopotamian flood story.

But older still than the Babylonian world map, and strictly practical, is a map from Egypt. Around 1150 BCE, in the reign of Ramesses the Fourth, a scribe named Amennakhte drew a map on papyrus for a stone-quarrying expedition. It covers about fifteen kilometers of a desert valley called the Wadi Hammamat, in Egypt's Eastern Desert. It shows the gold mines, the miners' settlement, the roads. And it colors the hills by rock type: black for one kind of rock, pink for another.

That makes the Turin Papyrus Map the oldest surviving geological map in the world. The next one we know of comes about twenty-nine centuries later.

Mesopotamia had been making practical plans even earlier: plans of fields, canals and property, and a city plan of Nippur from around 1500 BCE. The picture that emerges is that ordinary, workaday mapping of land and property came long before anyone tried to draw the whole world.

Which brings us to the Greek historian Herodotus, and his story of how surveying began. A king of Egypt, he says, divided the land into equal plots and taxed each one every year. When the Nile flooded and carried part of a plot away, royal surveyors came out to measure it again, so the tax could be adjusted. "From this practice, I think," Herodotus writes, "geometry first came to be known in Egypt, whence it passed into Greece."

Those surveyors had a wonderful name. They were the *harpedonaptai*, the rope-stretchers, because they worked with knotted, stretched cords. Whether or not Herodotus got the causation exactly right, Egyptian geometry really did grow out of measuring land on the floodplain. And the Great Pyramid, almost perfectly square and almost perfectly aligned north to south, tells you how good Egyptian surveyors already were, many centuries before he was writing.

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Step back, and there's a pattern here that will keep turning up for the rest of this series. Mapping moves fastest wherever the stakes are highest. Taxes on the Nile. Stone quarries in the desert. A canoe and an island you can't see yet. Later it'll be armies, empires and shipping. Maps appear wherever not knowing where things are costs the most.

But notice, too, what has quietly changed between the start of this episode and the end. The Marshallese navigator's map lived in his head; the stick chart usually stayed on the beach. The Egyptian surveyor's map lived in the king's tax records. Spatial knowledge has started to move out of people and into institutions: into archives, offices, the state. It's the first small step in a long story about who gets to hold the map.

Next time, the Greeks take Herodotus at his word. A librarian in Alexandria uses a stick and a shadow to measure the size of the entire planet. A brilliant astronomer makes a mistake that will help send Columbus west. And a carpenter-turned-clockmaker builds a watch that finally makes it possible to pin the whole Earth to a single grid.

I'm Paul. This has been *From Songlines to the Cloud*. Thanks for listening.

[Theme out]